



Fall 2021

Understanding Light Pollution in Puerto Rico: A Pathway to Guarantee Visual

Access to Our Galaxy

Abdiel R. Camacho Troche, Marinelys Cordero Rivera, Fabiana Henríquez, Nathan Millán,
and Danya S. Santiago

STAR Academy at The Arecibo Observatory

Abstract

Have you ever looked at the sky and wondered where are all the stars? If you watch movies or pictures on the internet, you can see a sky full of stars, but when you look up into the sky you may only see very few stars. Relatively recently, between the 19th and 20th centuries to be exact, light pollution started to become a problem. It has gotten worse with the passing of years, and its negative effects have become more significant. One of the many adverse effects that stood out to us, aspiring astronomers, was that light pollution impeded our visual access to the galaxy. This, for us, represented a genuine worry and was our inspiration for our research project. Our project's goal consisted of looking for a way to guarantee visual access to our galaxy. After extensive and deep research and analysis, we concluded that the best way to reduce this is by designing a light post model that could help us represent our idea of how a reconstructed light post could help us reduce light pollution. This light post describes the different mechanisms we want to add, like a reprocess of light, a warmer tone of light, and a position finder. To obtain the results of this project, the light that was scattered/spilled out of the concentrated area had to be measured using a photometer to then compare it with the data of a standard light post and prove its efficacy.

Keywords:

-Sky

-Excessive light

-Light post

-Visual access

-Features

1. **Introduction**

It is common to hear presentations about climate change, the effects of space junk, light pollution, etc. If we were to be carried away by the amount of information that pops up on the news and the internet, we would believe that these are topics dominated by the inhabitants of Earth, but no. We still must educate more about these topics to provide a broader understanding. For example, if we were to do a random survey, we would find that a significant part of the population does not know that light generated on Earth adversely affects space; and this is a problem that we need to spread awareness for.

So now you may be wondering, why is light pollution an interesting topic to develop anyway? Light pollution determines what we can explore in the night sky and we as a group believe light pollution can be eradicated, not only for our homes but for the whole planet. Friendly telescopes captivate less than a not-friendly telescope. Not only does it transmit problems into our galaxies but affects human and wildlife vision. “People are so sensitive to light pollution that even a full moon will disrupt our shut-eye” (Brookshire, 2019). This conflict brings blindness to people highly exposed to its efficacy, interrupts photosynthesis on plant species, and disrupts “night hunts” from nocturnal life.

Day after day new investigations are being conducted. But as research and new findings advance, the damage to the visual pathway and our home planet is exponentially growing. But all hope is not lost since this damage can be reversed. There are diverse civil, scientific, and research organizations that are in favor of ideas that can minimize the damage of current inventions that are fascinating to some, economically viable for others but are causing severe damage to the visual pathway to our galaxy and our quality of life. Only together, we can guarantee visual access to future astronomers and lovers of science, who can develop greater technologies that approach higher visuals into the night sky.

Methodology

2.1. Research Approach and Methods:

To achieve our goal of guaranteeing total visual access to our galaxy, we decided to create a model which represents: a reprocess of light, a warmer tone of light, and a position finder. The use of past experimentation and thoughts led us to collect all methods possible to design a new lamp post. Our investigation resulted in both numerical and standard qualities. Quantitative data led us to decide a height for our model and the light efficacy, and qualitative data for our sensor and reflective mirror. Our investigation results were gathered by comparing standard data from the used light post in Puerto Rico, scientific journals, and past

experimentation as expressed. To conclude, we decided to create a physical model which represents all factors expressed. Other ideas of creating a 3D model were discarded due to lack of time and less experience. We shared our areas of research and distributed them into shades, a sensor system, and a warm tone and shape for the lightbulb. The shades will redirect the light efficacy pointing only downwards, the sensor system will detect movement with an infrared detector, preventing the 24/7 waste of energy, and the warmth tone frequency will be 3,000 K units on LED lights. After analyzing the performance of used light posts in Puerto Rico areas, we decided that our model should be used in areas with urbanizations because its shade, sensor, and reflective mirror work best in this type of environment. All members shared thoughts and results from expressed areas, and after reviewing each other's parts, created the thought of making a physical model that could guarantee us future scientists and former astronomers, a better view of our galaxy. How could we prove its effectiveness? Nodaway light posts spread light energy sideways or upwards, but our light post will only spread light downwards with the use of a reflective mirror or shades, which prevents it from spreading sideways. Outdoor light uses 120 terawatt-hours of energy, while our light post has a 3,000 K frequency, will use LEDs and will perform a low efficiency of lighting, when no movement is detected. Using our sensor, if no movement is performed, light efficacy will be at a low capacity, but when movement is performed, its efficacy will be higher. All these factors contribute to the creation of a new light post, which can prevent criminality, car accidents, nonvisual skies, and disrupts the natural pattern of the environment.

2.2. Justification:

The methods that we design are with the purpose of reducing light pollution and guaranteeing visual access to our galaxy. One of our methods was to incorporate a sensor that when it detects movement the light will brighten up. This is going to help ensure that the light being emitted from the lamp post is not as much as normally it will, while still protecting the security of people. Another method is to use a light bulb of 3,000 K which is a much softer white. All these with the purpose to make sure we are only lighting the ground and are not reflecting other objects, redirecting the light where we don't need it. Our light post is going to have a shield that will help us to redirect that light, so we don't have unnecessary light spreading upwards. Another method that we are going to use is a reflective inside made of aluminum that will recycle the light and reflect it out again. We will also include a mechanism to the shields so they can have the ability to look up and down. This will help so we can redirect the light being emitted only where it is needed and avoid having to put more light posts.

2.3. Variables that will be taken into consideration:

The variables that are to be considered in the design of this light pole are as follows. The number of trees near the road, if the area is urban or rural, citizen's safety and criminality, the distance between the light poles, the funds that may be allocated to the development of the

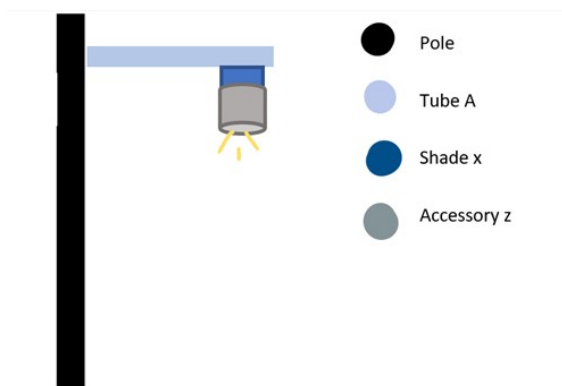
project, and maritime zones with endangered species. All of these variables are to be considered for the design of the project to ensure that it will complete its established objectives. Because if the area has a lot of tree branches near it will activate every time the wind blows the branches near the sensor. We also need to determine if the area is urban or rural to be able to establish the number of light poles that need to be installed because in a rural area we would need more light poles to guarantee a safe transit, but in an urban environment, we would require less light poles to be installed. We also need to find the cost that the project requires because we do not know the budget that could be allocated to design it and determine the amount that we may need to create. We need to do this to prove that it is cost-effective. Finally, we need to know if it is going to be installed in a maritime zone with endangered species to be able to use a dimmer light fixture to prevent the creature from becoming disorientated.

2.4. Steps and Materials:

Now, we will describe how the light pole design will be built and its exact measurements:

We will model a cylindrical-shaped light pole that's 4" in diameter and 4' with 4 ½" tall. This will have a 1' long and 6 ½" diameter connecting tube A (arm). The connecting tube (A) will be attached to the post at a height of 4' and 4 ½" off the ground. A 2 ¾" tall, 3 ½" diameter, cylinder-shaped PVC shade will also be made. The inside of which will be covered with reflective tape. There will be a ceiling rosette connected to a tube (A) fixed at about 4" from the end of the tube looking downward with cables connected to it that are about 10 ft

long, these cables will go through the tube and come out of the side of the pole. All the parts will be fastened to a 2'4 ½" x 11 3/8" x 5/8" wooden base. We will install a (type and color) lightbulb and create an aluminum accessory (z) in the shape of a cylinder 4" in diameter and 4 7/8" in height. This and the PVC shade (x) will be drilled (with a 1/8" bit) 7/8" apart between each piercing and leaving 3/8" from the top edge and 5/8" from the bottom. Finally, a 1/8" screw will be used to secure the accessory (z) to the shade (x) in the desired position.



2.5. Materials Table:

Materials	Service of the materials
PVC pipes	Create the posture
Metal sheet	Used to create the accessory
Reflective and electric tape	Adjust light efficacy and direction

Adjustment rings	Used to hold the accessory together
------------------	-------------------------------------

Ceiling Rosette	Power up the light bulb
-----------------	-------------------------

Incandescent bulb	Express light efficacy
-------------------	------------------------

Wires	Give power to the ceiling rosette
-------	-----------------------------------

Rubber bands	Prevents the shade from falling (for security)
--------------	--

Screws	Adjusts the accessory in place
--------	--------------------------------

Extension	Makes plugging in the lightbulb easier
-----------	--

Paint (optional)	Demonstrate creativity
------------------	------------------------

Tools	
--------------	--

Drill	
-------	--

Screwdriver	
-------------	--

Hacksaw	
---------	--

Photometer	
------------	--

2.6. Principal Obstacles:

At night, lighting enhances the perception of safety. It both allows pedestrians to see one another and prevent major accidents and, through the reduction of concealment in public, decreases crime activity. This is why one of the main obstacles we met with was favoring both citizens' security and the reduction of light pollution. This became a barrier/limit at first; however, through extensive research and teamwork, we were able to develop a feature that favors both aspects. First, we thought of using light timers in which, at late night hours, the lights would remain off. However, that would become both a complication and a threat for the citizens who drive during these hours for different reasons. Taking this into consideration, we then decided to change that feature into a motion sensor. However, to assure even more safety, we decided to implement a motion sensor that, instead of its constant being off and when motion is detected turning on, it's constant being a dim light, and through the detection of motion, the light would reach its total capacity. Furthermore, we also ran into another obstacle when starting the development of a complex animated graphic design of our lamppost model: the applications we found that allowed team collaboration were let alone complex without taking into consideration the process of animating the graphic. Because of this, we decided to make a physical model instead.

3. Results:

Our Research Project is about Understanding Light Pollution in Puerto Rico: A Pathway to Guarantee Visual Access to Our Galaxy. So, we needed to make a strategy to Guarantee Visual access to our galaxy and understand it better. For this task, we created a theoretical framework where we all took our time to investigate more about light pollution and put it in simple words. This work helps us to understand it better and plan a strategy so we could reduce all the light pollution in Puerto Rico. The strategy we developed together with a brainstorm of ideas was to make a new light post, but not a normal light post one that helps us reduce the light pollution without having to take the posts down. This light post has different types of mechanisms that will help us reduce light pollution. These mechanisms are a shade, a sensor, and a specific color temperature for the light bulb. In addition to that, we are going to put it in a location where the light bulb can be a better benefit for all. The real question is how we are going to prove that our light post is better and more beneficial than the normal one. For this task, we will use a photometer. With this instrument, we are going to measure all the data and compare it with the data of the normal light post.

3.1. Photos:

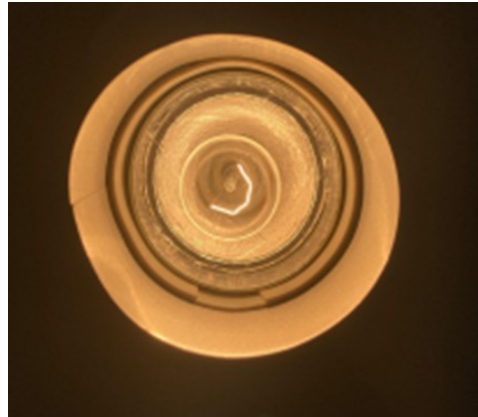
Side View of the light post: Light Post design (without shade): Taking the measurements:



Lit up view:



Inside view of the shade:



3.3. Color and temperature of the light bulb:

When comparing 4,000K lights with 3,000K lights we must know what they are. First, 4,000K lights are a bright white color that is used to imitate the morning and afternoon sun.

And 3,000K lights are a warm white which is a yellowish light that creates a more relaxed environment. We can observe that there is a 3.1% decrease in the emission of blue light in the 3,000K light which impacts the production of melatonin by 15.2%, while 4,000K lights impact it by 18.3%. Second, we see that there is a 3.1% decrease in the impact of melatonin production which makes the 3,000K light less harmful. And when comparing the impact on the night sky by 4,000K light we see that it affects the visibility of the night sky by 13.5% while 3,000K light only affects the visibility by 2.2% which is a decrease of about 11.3%.

3.4. Shade:

To be able to see how much light is being scattered/spilled out of the concentrated area of illumination, we measured the intensity of the scattered light at two different heights: 1'6" and 3'5" from the floor. The measurements that are 1'6" feet from the floor will be used to measure invasive light and the one that's 3' 5" ft from the floor will be used to measure the light that shines upward, contributing to skyglow. Both of them will use the same format to collect information: first, measuring the controlled variable (representing the common light post) light's intensity 1.5, 3, and 6 feet away from the light source; then doing the same with the two different accessory positions (1 and 2); the only difference between the intensity and skyglow measurements are the heights from where they will be taken. The results shown in the following charts and graphs help us understand how the three accessory positions reduce the scattered light.

Table 1C. Control Position, First Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.83	4.05
3	2.81	1.00
6	0.85	0.57

Table 2C. Control Position, Second Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.88	4.00
3	2.86	1.02
6	0.86	0.62

Table 3C. Control Position, Average

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.86	4.03
3	2.84	1.01
6	0.86	0.60

Table 1F. First position, First Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.00	1.62
3	2.03	0.69
6	0.58	0.42

Table 2F. First position, Second Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.25	1.60
3	2.10	0.71
6	0.58	0.42

Table 3F. First position, Average

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	6.13	1.61
3	2.07	0.70
6	0.58	0.42

Table 1S. Second position, First Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	4.53	1.15
3	1.84	0.60
6	0.46	0.36

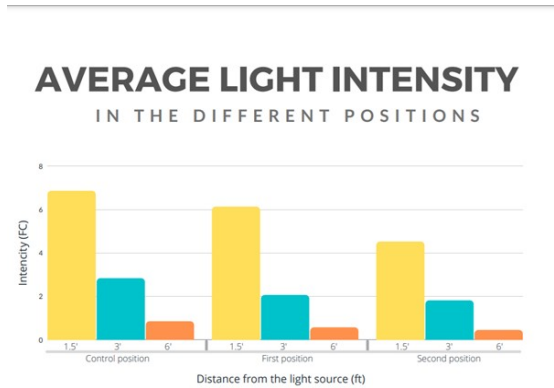
Table 2S. Second position, Second Trial

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5"height
1.5	4.52	1.10
3	1.80	0.58
6	0.47	0.35

Table 3S. Second position, Average

Distance from the Light source (ft)	Invasive Intensity (FC) at 1'6'' height	Skyglow Intensity (FC) at 3'5''height
1.5	4.52	1.13
3	1.82	0.59
6	0.46	0.36

Invasive:



Skyglow:

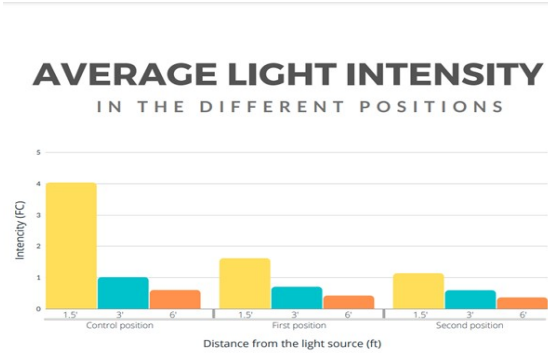


Table A. First position reduction in light intensity compared to the control position.

Distance from the Light source (ft)	Invasive Intensity (%) at 1'6'' height	Skyglow Intensity (%) at 3'5''height
1.5	11	60

3	27	31
6	33	30

Table B. Second position reduction in light intensity compared to the control position.

Distance from the Light source (ft)	Invasive Intensity (%) at 1'6'' height	Skyglow Intensity (%) at 3'5"height
1.5	34	72
3	36	42
6	46	40

3.5. Location:

Our mission is to guarantee visual access to our galaxy using an improved light post, which contains: a sensor that can identify or detect movement, a shade that can prevents unnecessary light going outwards, a light bulb that can project a certain smaller number of lumens and

watts, and an area where our vision could be achieved. We concluded that our light post should have access to municipalities with urbanized areas. Urbanizations are composed of developing cities, which reflects the relation between city dwellers and the total census of citizens on a state or an unincorporated territory. Overall, more citizens spend their lives in urban areas, which is why we think our light post will have a greater impact on these regions. Our sensor will identify any movement and project a higher luminosity either for nearby automobiles, jogging, or biking athletes. However, we believe that habitats and plant species should develop without any negative impacts. The implementation of a shade mechanism will only redirect the illumination downwards and project a lower dim when no movement is encountered. In other words, our energy source will present a lower percentage of its capacity when no movement is disclosed. And to complete our presentation of an improved light post, we will add a light bulb that can project lower watts when it is not in use and a higher lumen when it is instructed by the sensor mechanism. Now, why would all these factors outright our expectations on guaranteeing visual access to the sky? The difficulty of studying or even admiring the stars was indeed a 'biggy', but we compared data with the commonly used light post, how much energy is used, how outdoor lighting provokes sky glare, and so much more. We want our plant species to complete photosynthesis, the development of wildlife, and enjoy what our ancestors considered once as fascinating. Due to our plans for future inquiry, we cannot offer exact data, otherwise preliminary results. Also considering that our light post will have no effect during the daytime, we cannot guarantee this will reduce light pollution.

Having our Sun as the primary source of light, our light post energy source will have no use during the day, otherwise during the night. Being close to winter, it is tipped that clear nights are approaching, and it will be a good time for stargazing. We believe our light post will help many people, as well as prevent automobile accidents, loss of vision, prevent trees from bringing out buds earlier and bring forth an early spring.

4. Discussion

4.1. Materials:

Our eyes lack vision; they can no longer recognize other mysteries outside the Exosphere. We proposed the idea of recreating a light post, but with less radiant fabric. Our light post or lamppost will be built with the following: two PVC pipes, one metal sheet and a reflective tape, adjustment rings, one ceiling rosette, a set of wires, screws, and rubber bands. A cylindrical-shaped pole with a 4" diameter and 4' with 4" and a half tall, will connect to the one 6 and a half inches PVC tube. Being already attached, it will connect to the other 4' and 4" and a half PVC tube above the ground. The second PVC tube will connect the 2" and three-quarters tall and 3" and half in diameter shade, being covered on the inside with reflective tape. To unite both PVC tubes, a ceiling rosette about 4" will allow a 10 ft pair of wires to transfer between both PVC tubes, in order to give energy to our LED bulb. As a base, a wooden piece of 2 '4 and a half inch's times 11 three eight times five-eight. A specialized LED bulb will portray the emitted energy with the use of a 4" diameter aluminum cylinder seven-eighths tall. Both aluminum cylinder and reflective mirror will prevent the emission of

excessive energy being reflected side-to-side or upwards. Both pieces will be instructed with a seven-eighths separation space with each piercing and only clear out three eighths from the peak and five eighths from the bottommost. To conclude, a one-eighth bolt or screw will tight the aluminum cylinder to the shade, only portraying the emitted light downwards. Being reminded that this is just our set of ideas on a mockup, but a light post was used as a reference. Each piece represents a fundamental purpose in our light post. Our base will support and adjust the established height and both PVC tubes will hold on to the wires, the bulb, the aluminum cylinder, and the shade mechanism. Also, the use of a drill, a hacksaw, and a screwdriver made possible the installment of our project. The idea of installing a movement reflector or sensor has not yet been installed in our post but if we manage to accomplish another one of our goals, then our post could transmit a lower emission of energy when no movement is detected and a higher emission when movement is perceived. We believe that our model could demonstrate or visualize how modern-day light posts should collaborate and why future generations will miss out on what our ancestors once enjoyed. If light pollution continues having an impact, engineers might consider the use of expensive technology, which could only benefit their studies and not our pleasures of enjoying a clear sky.

4.3. Color and temperature of the light bulb:

When comparing 4,000K lights with 3,500K lights we must know what they are. First, 4,000K lights are a bright white color that is used to imitate the morning and afternoon sun. And 3,500K lights are a warm white which is a yellowish light that creates a more relaxed

environment. We can observe that there is a 3.1% decrease in the emission of blue light in the 3,500K light which impacts the production of melatonin by 15.2%. While 4,000K lights impact it by 18.3%. Second, we see that there is a 3.1% decrease in the impact of melatonin production which makes the 3,000K light less harmful. And when comparing the impact on the night sky by 4,000K light we see that it affects the visibility of the night sky by 13.5% while 3,000K light only affects the visibility by 2.2% which is a decrease of about 11.3%. From the use of Light color temperature of 3,500K, we observe that the impact on wildlife, humans, and the visual pathway to our galaxy are affected in a less severe way, therefore, reducing the light pollution more when in combination with light shading when it is a 4,000K light color temperature without light shading. This is because 3,500 K light color temperature is an imitation of the morning and afternoon sun that is a yellow light with a “true to life” color.

4.4. The shade:

To determine which shade to use for the design of our light post, we took into consideration a series of shades that would fit our vision. It was clear that we needed to use a shade that didn't let light escape above, spreading light into the sky, but there are many shades with different shapes that accomplish this. Of course, the first ones to go were the ones that pointed upward or had their sides completely exposed, these directly lit up the sky and didn't light up below it where it's needed. Then, we discarded the shade that only covered part of the light source because even though it is a better option than the previous one, looking directly at the

light source is not favorable and causes glaring. After that, we took a look at the cone and cylindrical shaped shades (that covered the light source completely from the top and sides), neither had anything particularly different or bad so we could pick either one of them, and we chose to go with the cylindrical one because of the minimalism we planned on it having.

The shade system we chose to install decreases the use of unnecessary light and can be accustomed to the area it is located from the controlled position (the shade has “3” positions: Control, 1, and 2). The system lets you move part of the shade up and down to cover the area needed or desired. For the first position, you move the accessory down vertically $7/8$ ” to the second set of holes (from the control position); and the second position is $7/8$ ” below, where the third set of holes are located.

We made sure that the photometer was calibrated before this experiment. Since we were going to use a 40-watt lightbulb to test it, we looked for the conversion of 40 watts to lumens which resulted in 450 lumens. Then we looked up how many lumens is 1FC, which was 10.76 lumens. Divide 450 lumen by 10.76 lumens to get the average amount of CF in a 40-watt bulb. This mathematical operation gives the theoretical result of 41.82CF, that is, 42FC. When we measured with the instrument directly at a distance of 12”, we obtained an experimental reading of 45FC. When looking for the percentage of error, we take the theoretical result, subtract it from the experimental reading, divide the result by the theoretical result and multiply it by one hundred; we obtained 7.14%. In general, a reasonable

error percentage ranges from 5% to -10%. We understand that this percentage of error is reasonable and therefore, conclude that the instrument is calibrated.

$$FC_{40 \text{ watts}} = \frac{450 \text{ lumens}}{10.76 \text{ lumens}} \quad (1)$$

When looking at the measurements, we have registered a drastic drop in light intensity from the control position to the first position, specifically at the distance of 1.5' and at a height of 3'5". We recognize that this drop in intensity is attributed to the effect that the shade's accessory makes at that elevation and distance, different from the other 3' and 6' distances, where the effect of the shade's accessory does not prevent a regular amount of intensity from arriving. So, in order to have a general conclusion on how much the accessory reduces the amount of light going outwards, we won't be using the measurements taken 1.5' from the light source. This does not mean that we will discard these measurements however, it just would not be accurate to say that the mechanism reduces 60% of the light going outwards in general.

The shade shadow effect that we found in the measurements at skyglow height (1.5' away from the light source) we can also find manifested (to a smaller degree) at 3' away from the light source at skyglow height. When we compare the light's intensity at 3 and 6 feet, we can

see that they have similar reduction percentage values. We understand that the further away from the light source, there should be a greater reduction in intensity, but the reductions are similar. But because the shade's shadow effect affects the intensity of the light drastically at 1.5', it also has an effect when measured at 3' but it's more subtle. This is what makes the 3' intensity percentages very similar to the 6' intensity percentages.

Also, the intensity of the light from position 1 measured 1.5' away from the light source for invasive light (1'6") is affected because the light is being focused into a concentrated area. When the shade is in its first position, it only saves 11% of the light compared to the controlled position because the light is concentrated close to it.

Having this in mind, these are the percentages of light intensity reduced by the shade's positions:

- In the shade's first position, when measured at skyglow height (3'5"), it showed an average of a 30% reduction in intensity; and when measured at invasive height (1'6"), showed a reduction ranging from 11%-33%.
- In the shade's second position, when measured at skyglow height (3'5"), it showed an average of a 40% reduction in intensity; and when measured at invasive height (1'6"), showed a reduction ranging from 34%-46%.

This mechanism idea came to be by observing an adjustable flashlight and as mentioned before, its main purpose is to reduce unnecessary lit-up areas. An example of this is a light

post between two properties. If it lights up too much and makes looking out the window impossible, then the owner could move it to the first position and if the property that's next to it still wants less light, the owner could lower the shade down to the second position, so it's concentrated in his area. Another example could be that someone just wants a little light on their patio so they can go read in the afternoons but has gatherings there the weekends, so he lifts the shade, so the place is better lit. This feature was designed to be as convenient to the user and good for visual access to our galaxy as possible.

5. Conclusion

Light pollution is all the artificial light disturbing the natural light of our sky. Why does this occur? This happens because of the excessive and poor design of outdoor lighting. It is important to take care of this problem because this is not only a problem that affects the vision of our sky. It affects the civilian's health because it eliminates the melatonin that it's a very important hormone, it disturbs the cardiac rhythm and affects our natural sleep cycles and it elevates rates of diabetes, obesity, breast and prostate cancer. Not only humans get affected by this problem but also wildlife. This happens because a lot of diurnal animals need the night to get feed, mating, and keep their normal migration patterns (Lowenthal, 2021). It is important to keep our animal and human life safe and for this, my team and I wanted to look for a solution. Not only to keep us and animals safe but, we want a nice clear night sky that is not polluted with artificial light. We want to look upwards and see a world beyond ours,

something new that we can see a sky full of stars and mysteries. Our main goal is to guarantee visual access to our galaxy. To do this we need to know the principal causes of light pollution which are: light trespassing, glare, and sky glow. The street light causes a lot of glare which is one of the main causes of light pollution. With all the information gathered we decided that one of the main options was to design a lamp post that would help reduce light pollution. To create this lamp post we had to view all the details to create a post that helps reduce light pollution without putting the life of civilians in danger. To achieve this we need to create different types of mechanisms and pay attention to the small details that could make the difference in the light post. Our light post will have a motion sensor. Our motion sensor will work so that when motion is detected, a specific action will be conducted. This motion sensor will not be like a standard motion sensor in that the action is the light turning on and then turning off. Our motion sensor's action will be to increase the intensity of the light in brightness after prevailing slightly dim. The color temperature of the light bulb is going to be 3,000k. This light is a warm white and a yellowish light which creates a more relaxed environment. This type of light only affects visibility by 2.2%. We also are going to include a type of shade that it's going to have a reflective inside that is going to help us redirect the light so that we don't have unnecessary light reflecting on other surfaces. This shade will have the ability to go up (first position) and down (second position) so that it could be adjusted to the light that is necessary for that area. To measure how much light is being scattered/spilled we need to measure them in two different heights: 1'6'' and 3'5'' ft from the floor. The amount of

light being saved in “first position” is 30% while the amount of light being saved in “second position” is 34%-46%. The measurements are going to be used to measure the invasive light and the skyglow. To prove our goal to guarantee total visual access to our galaxy, we decided to create a model which represents: a reprocess of light, a warmer tone of light, and a position finder. We have achieved our goal because with all the data gathered we have proven that it was more effective. The use of past experimentation and thoughts led us to collect all methods possible to create a new lamp post. Our investigation resulted in both numerical and standard qualities. Quantitative data led us to decide a height for our model and the light efficacy, and qualitative data for our sensor and reflective mirror. Our investigation results were gathered by comparing standard data from the used light post in Puerto Rico, scientific journals, and past experimentation as expressed.

Acknowledgments

We would like to thank our STAR professors, Zaian Lopez, Yazmin Santiago, and Bradley Rivera for their help and guidance, without them we wouldn't have been able to complete such a project, as well as to all the parents behind the team that supported and pushed us to do our best every step of the way.

References:

5 benefits of having a motion sensor installed on your property. Beacon Home Services.

(2018, January 19). Retrieved October 5, 2021, from

<https://www.beaconelectricalservices.com/blog/2018/january/five-benefits-of-having-a-motion-sensor-installe/>.

Brady, H. (2021, May 4). *These Energy-Saving Bulbs Are Making One Pollutant Much Worse.* Science.

<https://www.nationalgeographic.com/science/article/light-pollution-energy-LED-bulbs-spd?loggedin=true>

Brookshire, B. (2019, December 6). *Scientists Say: Light pollution.* Science News for Students.

<https://www.sciencenewsforstudents.org/article/scientists-say-light-pollution>

Buda Texas, Official site (s.f.). *How you can help preserve the night sky.* Buda Texas Government. <https://www.ci.buda.tx.us/656/How-You-Can-Help-Preserve-the-Night-Sky>

Caravoc M. (November 30, 2015). *How does color temperature affect light pollution.* Wyoming Stargazing Organization.

<https://www.wyomingstargazing.org/how-color-temperature-affects-light-pollution/#>

Cinzano, P., & Falchi, F. (2012). The propagation of light pollution in the atmosphere. *Monthly Notices of the Royal Astronomical Society*, 427(4), 3337–3357. <https://doi.org/10.1111/j.1365-2966.2012.21884.x>

De La Cruz, B. (2019, July 30). *How to add a motion sensor to existing outdoor lights*.

Home Security & Smart Home Reviews. Retrieved October 9, 2021, from <https://homealarmreport.com/smart-home/how-to-add-a-motion-sensor-to-existing-outdoor-lights/>.

Dark Sky | *Light Bulb Types*. (n.d.). Bulbs.Com. Retrieved October 5, 2021, from <https://www.bulbs.com/learning/darksky.aspx>

Definition of IR detector. PCMAG. (n.d.). Retrieved October 5, 2021, from <https://www.pcmag.com/encyclopedia/term/ir-detector>.

Drake, N. (2021, May 3). *Light pollution is getting worse, and Earth is paying the price*. Science.

<https://www.nationalgeographic.com/science/article/nights-are-getting-brighter-earth-paying-the-price-light-pollution-dark-skies>

Hill Country Alliance. (s.f.). *Night sky friendly lighting*. HCA.

<https://www.hillcountryalliance.org/nightskylighting#1594241452110-9bd8b523-eec3>

Hölker, F., Moss, T., Griefahn, B., Kloas, W., Voigt, C. C., Henckel, D., Hänel, A., Kappeler, P. M., Völker, S., Schwoppe, A., Franke, S., Uhrlandt, D., Fischer, J., Klenke, R., Wolter, C., & Tockner, K. (2010). The Dark Side of Light: A Transdisciplinary Research Agenda for Light Pollution Policy. *Ecology and Society*, 15(4). <https://doi.org/10.5751/es-03685-150413>

How to choose and install motion sensor lighting. Family Handyman. (2020, July

Retrieved October 5, 2021, from

<https://www.familyhandyman.com/project/how-to-choose-and-install-motion-detector-lighting/>.

Hudson, A. H. (2017, March 15). *Stargazing in the Outdoors: Getting Away From Light Pollution*. FIX.

<https://www.fix.com/blog/avoiding-light-pollution/>

International Dark-Sky Association. (s.f.). *Light pollution solutions*. IDA.

<https://www.darksky.org/light-pollution/light-pollution-solutions/>

International Dark-Sky Association. (s.f.). *Outdoor lighting basics*. IDA

<https://www.darksky.org/our-work/lighting/lighting-for-citizens/lighting-basics/>

James Lowenthal(n.d.). Light Pollution.

<http://www.science.smith.edu/~jlowenth/lightpollution/lightpollution.html>

LEC Expert blog - Thoughts and expert solutions in LEDs. (n.d.). Light Pollution: The Best Solutions for Outdoor Lighting.

<https://www.lec-expert.com/topics/light-pollution--the-best-solutions-for-outdoor-lighting>

Motion Light Sensors. Thomasnet. (n.d.). Retrieved October 5, 2021, from

<https://www.thomasnet.com/articles/instruments-controls/light-motion-sensors/>

Smart Street Lighting System: Motion sensor street lighting. Citintelly. (n.d.). Retrieved

October 9, 2021, from <https://www.citintelly.com/intelligent-street-lighting-products/motion-sensor-street-lighting/>.

Staff, L. N. (2020, August 27). *4 Types of Light Bulbs: Which is More Energy Efficient*. Lumen Now.

<https://lumennow.org/choosing-a-bulb/>

Town of Andover, Massachusetts. (s.f.). *Energy management. LED streetlight retrofit.*

<https://andoverma.gov/Faq.aspx?QID=224>

VOLT (n.d.). Using Dark-Sky Approved Lighting To Reduce Light Pollution.

<https://www.voltlighting.com/learn/using-dark-sky-approved-lighting-to-reduce-light-pollution>

Xuan-Hao Lee, Ivan Moreno, and Ching-Cherng Sun. (2013). High-performance LED street lighting using microlens arrays. Optica Publishing Group, 21(9), 10612-10621. <https://www.osapublishing.org/oe/fulltext.cfm?uri=oe-21-9-10612&id=253009###>

Cinzano, P., & Falchi, F. (2012). The propagation of light pollution in the atmosphere. Monthly Notices of the Royal Astronomical Society, 427(4), 3337–3357. <https://doi.org/10.1111/j.1365-2966.2012.21884.x>